

CAKE Web App Guidance

Guidance for how to use the CAKE Web App is given below (under Guidance).

Essential items are marked in **red** and with an asterisk*. Sublisted items will not be visible unless the parent item is selected. *Italicized* items may be inactive or invisible depending on other parameters.

Any time, mole, volume, and path length units can be used. However, units must be consistent between inputs, including upload data. Common examples of units are shown below.

Unit	Example
Time_unit	s, min, h
Moles_unit	mol, mmol, μmol
Volume_unit	mL, L, cm^3 , dm^3

An example of CAKE Web App use is described below (under Demonstrations).

Guidance

First, one must fill in the parameters as described below.

*Upload

Upload: upload of reaction data.

- **Select Data**: uploads calibration data in Excel format (.xlsx). The first row is assumed to be headings, i.e., only data from the second row onward are used for fitting. Number of rows must be equal for all data.
- **Sheet Name**: specifies the sheet name. Default is Sheet1.
- **Time Column**: specifies the location of the time column. The first column is 1. **Time column unit**: time_unit.
- **Simulate Reaction**: simulate the reaction. Default is to fit (not simulate).
- **Simulation Start**: start time of stimulation. Default is 0. **Unit**: time_unit.
- **Simulation End**: end time of stimulation. **Unit**: time_unit.
- **Simulation Time Interval**: time intervals of simulation. **Unit**: time_unit.

*System

System: reaction conditions.

- ***Initial Reaction Volume**: specifies initial volume of the reaction solution. **Unit**: volume_unit.
- **Continuous Volume Loss Rate**: specifies loss of monitored solution over time, for example into an instrument. **Unit**: volume_unit time_unit⁻¹. Default: 0 (no loss).
- **Est Rate Constant**: estimated overall rate constant of reaction. Default is “estimate” which determines an initial estimate of 10^x , where x is an integer between -13 and 13 concentration_unit time_unit⁻¹. Using application estimates slows down the fitting program significantly, especially for “Arrhenius” and “Eyring” equation fitting (see below).
- **Min Rate Constant**: minimum allowed overall rate constant of reaction. Default is “estimate $\times$ 0.001”, i.e., the estimated overall rate constant $\times$ 0.001.
- **Max Rate Constant**: maximum allowed overall rate constant of reaction. Default is “estimate $\times$ 1000”, i.e., the estimated overall rate constant $\times$ 1000.
- **Est Second Constant**: estimated second overall constant of reaction. Default is “estimate”. Using application estimates slows down the fitting program significantly, especially for “Arrhenius” and “Eyring” equation fitting (see below).
- **Min Second Constant**: minimum allowed second overall constant of reaction.
- **Max Second Constant**: maximum allowed second overall constant of reaction.
- **+ Species**: adds additional species.

- – **Species:** removes additional species.
- **For each species:**
 - **Use for Fitting:** specifies if the species should be used for fitting.
 - **Species Name:** specifies species name.
 - ***Species Type:** specifies the type of species: Reactant, Product, or Catalyst. Default is Reactant.
 - ***Stoichiometry:** specifies the stoichiometric coefficient of the species. Default is 1.
 - **Initial Moles:** specifies initial moles of species in reaction solution. **Unit:** moles_unit.
 - **Final Moles:** specifies final moles of species in reaction solution. Only required if importing intensity (not concentration) data. **Unit:** moles_unit.
 - **Column:** specifies the location of the species concentration or intensity column. Must be complete if “Use for Fitting” is selected. The first column is 1. **Column unit:** conc_unit or intensity_unit.
 - **Specify Order Limits:**
 - **Est Order:** specifies exact species order if simulating or “Min Order” and “Max Order” (see below) are left empty. Specifies initial estimate for fitting of species order if “Min Order” and “Max Order” (see below) are used. Default is 1.
 - **Min Order:** specifies the lower bound for fitting of species order. Fixes species order value if left empty. Default is 0.
 - **Max Order:** specifies the upper bound for fitting of species order. Fixes species order value if left empty. Default is 2.
 - **Continuous Addition:** specifies continuous addition of a solution.
 - **+ Addition:** adds additional continuous addition.
 - **– Addition:** removes additional continuous addition.
 - **For each addition:**
 - **Addition Solution Conc.:** specifies concentration of the solution being continuously added. **Unit:** moles_unit volume_unit⁻¹.
 - **Rate of Addition:** specifies rate at which solution is continuously added. **Unit:** volume_unit time_unit⁻¹.
 - **After Time:** specifies time at which continuous addition commences. **Unit:** time_unit. Default: 0, i.e., continuous addition commences immediately.
 - **Discrete Addition:** specifies discrete addition of a solution.
 - **+ Addition:** adds additional discrete addition.
 - **– Addition:** removes additional discrete addition.
 - **For each addition:**
 - **Addition Solution Conc.:** specifies concentration of the solution being discretely added. **Unit:** moles_unit volume_unit⁻¹.
 - **Volume Added:** specifies volume of the discretely added solution. **Unit:** volume_unit.
 - **At Time:** specifies time at which discrete addition occurs. **Unit:** time_unit. Default: 0, i.e., discrete addition occurs immediately.
- **Temperature Alteration:**
 - **Initial Temperature:** specifies initial temperature of reaction solution. Not required if temperature column is used. **Unit:** K. Default: 293.15.
 - **Temperature Gradient:** specified rate of temperature increase. Not required if temperature column is used. **Unit:** K time_unit⁻¹. Default: 0.
 - **At Time:** specifies time at which temperature gradient commences. Not required if temperature column is used. **Unit:** time_unit. Default: 0, i.e., temperature alteration commences immediately.
 - **Temperature Column:** specifies monitored temperature column. The first column is 1.
- **Discrete Subtraction:** specifies discrete subtraction of the solution, for example for removing aliquots.
 - **+ Subtraction:** adds additional discrete subtraction.

- – **Subtraction**: removes additional discrete subtraction.
- **For each subtraction**:
 - **Volume Removed**: specifies volume of the discretely subtracted solution. **Unit**: volume_unit.
 - **At Time**: specifies time at which discrete subtracted occurs. **Unit**: time_unit.

Fitting Options

Fitting options: equation fitting options.

- **Rate Equation**: specifies equation for simulating/fitting. Options are: standard, Arrhenius, and Eyring. Default: “Standard”. The equations are listed below, where k is the rate constant in the rate equation and the 1st and 2nd constants are the “Est Constants” under the “System” tab:
 - Standard: $k = k$, where **k is the rate constant** (1st constant).
 - Arrhenius: $k = Ae^{-\frac{E_a}{RT}}$, where **A is the pre-exponential constant** (1st constant), **E_a is activation energy** (2nd constant, in J mol⁻¹), R is the molar gas constant and T is temperature.
 - Eyring: $k = \frac{\kappa k_B T}{h} e^{\frac{\Delta S}{R}} e^{-\frac{\Delta H}{RT}}$, where κ is the transmission constant (assumed to be 1), k_B is the Boltzmann constant, T is temperature, **ΔS is entropy of activation** (1st constant, in J K⁻¹ mol⁻¹), **ΔH is enthalpy of activation** (2nd constant, in J mol⁻¹), h is the Planck constant, and R is the molar gas constant.
 - “Arrhenius” and “Eyring” equation fitting take significantly longer than “Standard” fitting.
- **Rate Algorithm**: specifies algorithm for solving initial value problems. Default: “Radau”.
- **Relative Tolerance**: specifies relative tolerance, i.e., number of correct digits, for solving initial value problems. Default: 10⁻⁶. Increasing this value accelerates solving but decreases accuracy and vice versa.
- **Absolute Tolerance**: specifies absolute tolerance, i.e., number of correct decimal places, for solving initial value problems. Default: 10⁻⁹. Increasing this value accelerates solving but decreases accuracy and vice versa.

Data Manipulation

Data manipulation: means to manipulate the reaction data.

Especially useful for noisy data and data not given as concentrations, e.g. data given in intensities from UV-Vis spectroscopy, NMR spectroscopy, IR spectroscopy, MS, etc.

- **Average Points**: specifies number of intensity data points to average across for scaling to species concentration. Default: 0, i.e., no scaling.
- **Smoothing Window**: specifies number of data points for rolling average. Default: 1, i.e., no smoothing.
- **Interpolation Multiplier**: specifies fraction of time points used for more fewer data or more accurate simulation. For example, 0.5 specifies using half the data while 2 specified using twice as many intervals. Default: 1, i.e., time points provided in “Time Column” are used for simulation.
- **Noise Fraction**: specifies fraction of noise relative to the initial concentration of the first species. Default: 0, i.e., no noise.

Output

- Once the above parameters have been filled in, the data can be fitted and/or downloaded, using “Fit” or “Download Fit” (see below). This changes the tab to the “Output” tab, eventually revealing results.
- **Simulate/Fit**: yields results online, shown as fitted parameters and figures of the obtained fits. These figures can be downloaded in SVG, PNG, EPS and PDF formats using the drop down format button and download button found below the figures.

CAKE Web App Guidance

- **Download Simulation/Fit:** downloads the obtained fit data (under FitData tab), the obtained fit parameters (under Outputs tab) and all the inputted parameters (under InputParams tab). This allows for easy tabulation of results and plotting of data in the desired software.
- Fitting typically takes ~5-60 s, depending on the amount of data, chosen fitting parameters and local internet speed.

CAKE Web App Guidance

Demonstration

An example simulation and fitting are shown for a reaction and experiment below. The simulated and fitted reaction data are available as “CAKE_example_sim.xlsx” and “CAKE_example_fit.xlsx” respectively.

A reaction between reactants “Reactant 1” and “Reactant 2”, forming a product “Product”, and catalysed by a catalyst “Catalyst”, has the overall equation:



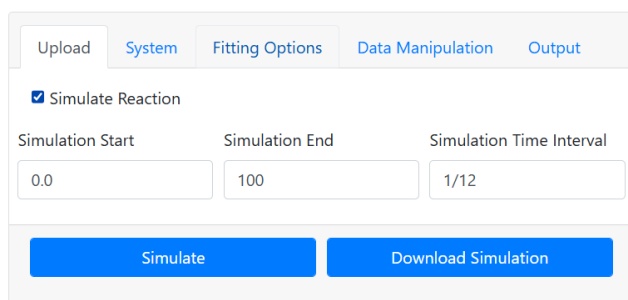
The reaction has a rate constant of $0.003 \text{ mmol L}^{-1} \text{ min}^{-1}$ and is first, zeroth, and first orders in “Reactant 1”, “Reactant 2”, and “Catalyst” respectively.

Simulation

First the reaction was simulated. These data were input into the web application as shown below. Units are shown in brackets for clarity but were not input. Parameters not mentioned were left as their default values.

Upload

- **Simulate Reaction:** ✓.
- **Simulation Start:** 0.0 (min, default).
- **Simulation End:** 100 (min).
- **Simulation Start:** 1/12 (min). I.e., one reading every 5 s.



System

- **Initial Reaction Volume:** 0.05 (L).
- **Est Rate Constant:** $0.003 \text{ (mmol L}^{-1} \text{ min}^{-1})$. Rate constant of reaction is $0.003 \text{ mmol L}^{-1} \text{ min}^{-1}$.
- **+ Species:** ×3 (4 species total).
- **Species Conditions:**

Species Name	Species Type	Stoichiometry	Initial Moles (mmol)	Est Order
Reactant 1	Reactant	1	10	1
Reactant 2	Reactant	1	20	0
Catalyst	Catalyst		0	1
Product	Product	1	0	0

- For “Catalyst” only:
 - **+ Continuous Addition:** ×1 (1 continuous addition).
 - **Addition Solution Conc.:** 1000 (mmol L^{-1}). Concentration of catalyst solution being continuously added is 1000 mmol L^{-1} .
 - **Rate of Addition:** $50\text{E-6 (L min}^{-1})$. Rate of continuous addition of catalyst solution is $50 \times 10^6 \text{ L min}^{-1}$.
 - **After Time:** 5 (min). Continuous addition of catalyst solution commenced at 5 min.

CAKE Web App Guidance

Upload

System

Fitting Options

Data Manipulation

Output

Initial Reaction Volume

Continuous Volume Loss Rate

0.05

0.0

Est Rate Constant

Min Rate Constant

Max Rate Constant

0.003

Estimate * 1E-3

Estimate * 1E3

Reactant 1

Reactant 2

Catalyst

Product

☐ Use For Fitting

Species Name

Species Type

Stoichiometry

Catalyst

Catalyst

1

Initial Moles

Final Moles

Column

0

Specify Order Limits

Specify Poisoning Limits

Est Order

Min Order

Max Order

1

0

2

Continuous Addition

Discrete Addition

Addition Solution Conc.

Rate of Addition

After Time

1000

50E-6

5

+ Addition

- Addition

+ Species

- Species

Temperature Alteration

Discrete Subtraction

Simulate

Download Simulation

Data Manipulation

- Noise Addition:** 0.02. Noise will be 2% of initial Reactant 1 intensity.

Upload

System

Fitting Options

Data Manipulation

Output

Interpolation Multiplier

Noise Addition

1

0.02

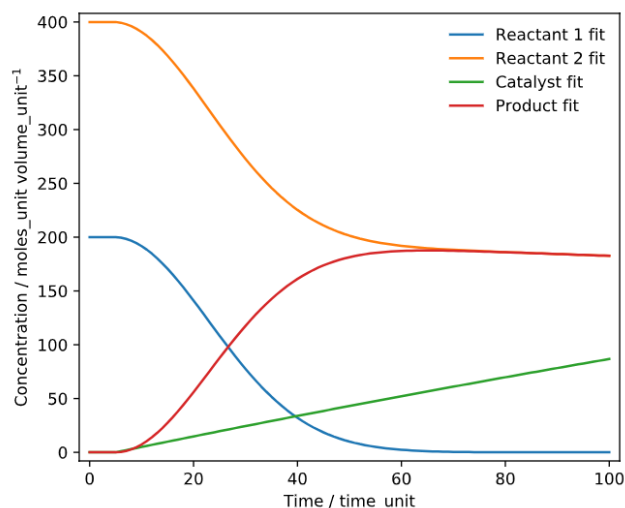
Simulate

Download Simulation

Output

Simulate:

CAKE Web App Guidance



Download Simulation:

These results are available as “CAKE_example_sim.xlsx”.

- FitData:

	A	B	C	D	E
1	Time / time_unit	Reactant 1 fit conc. / moles_unit_volume_unit^-1	Reactant 2 fit conc. / moles_unit_volume_unit^-1	Catalyst fit conc. / moles_unit_volume_unit^-1	Product fit conc. / moles_unit_volume_unit^-1
2	0	200	400	0	0
3	1	200	400	0	0
4	2	200	400	0	0
5	3	200	400	0	0
6	4	200	400	0	0
7	5	200	400	0	0
8	6	199.5009235	399.3011233	0.999000999	0.299276315
9	7	198.4083642	398.0091626	1.996007984	1.192434224
10	8	196.7332591	396.1350537	2.991026919	2.668535541
11	9	194.4916356	393.6948228	3.984063745	4.711551686
12	10	191.704353	390.7093281	4.975124378	7.300622161
13	11	188.3967923	387.2039493	5.964214712	10.41036479
14	12	184.5984565	383.2081884	6.951340616	14.01127538
15	13	180.3425563	378.7552548	7.936507937	18.07014207
16	14	175.6655196	373.8815751	8.919722498	22.5505359
17	15	170.6064749	368.6262769	9.900990099	27.4133271
18	16	165.2067222	363.0306589	10.88031652	32.61721452
19	17	159.5091687	357.1376272	11.85770751	38.11928983
20	18	153.5577832	350.9911494	12.83316881	43.87558307
21	19	147.3970392	344.6356979	13.80670611	49.84161961
22	20	141.0714255	338.1157605	14.77832512	55.97290949
23	21	134.6247733	331.475167	15.7480315	62.22562044
24	22	128.1000306	324.7568644	16.71583088	68.55680325
25	23	121.5385976	318.0022518	17.68172888	74.92505661

The simulation has the classic S-shaped curve associated with CAKE experiments of systems with a first order reactant and catalyst.

Fitting

The simulated reaction was then fit. These data were input into the web application as shown below. Units are shown in brackets for clarity but were not input. Parameters not mentioned were left as their default values.

Upload

- **Select Data:** CAKE_example_sim.xlsx.
- **Sheet Name:** FitData.
- **Time Column:** 1.

CAKE Web App Guidance

Upload

System

Fitting Options

Data Manipulation

Output

Select Data

Sheet Name

Time Column

Choose file

CAKE_exa..._data_2.xlsx

FitData

1

☐ Simulate Reaction

Fit

Download Fit

System

- **Initial Reaction Volume:** 0.05 (L).
- **Est Rate Constant:** “Estimate” ($\text{mmol L}^{-1} \text{ min}^{-1}$, default). The code will approximately estimate an initial rate constant for fitting.
- **Min Rate Constant:** “Estimate * 0.001” ($\text{mmol L}^{-1} \text{ min}^{-1}$, default). The code will fit the rate constant to a minimum of $0.001 \times$ the above initial estimated rate constant.
- **Est Rate Constant:** “Estimate * 1000” ($\text{mmol L}^{-1} \text{ min}^{-1}$, default). The code will fit the rate constant to a maximum of $1000 \times$ the above initial estimated rate constant.
- **+ Species:** $\times 3$ (4 species total).
- For “Reactant 1” only:
 - **Use for Fitting:** $\checkmark$. We will assume that “Reactant 1” concentration data was collected by our instrument.
- Species Conditions:

Species Name	Species Type	Stoichiometry	Initial Moles (mmol)	Column (column in mmol L^{-1})	Est Order	Min Order	Max Order
Reactant 1	Reactant	1	10	2	1	0	2
Reactant 2	Reactant	1	20		0		
Catalyst	Catalyst		0		1	0	2
Product	Product	1	0		0		
- For “Catalyst” only:
 - **+ Continuous Addition:** $\times 1$ (1 continuous addition).
 - **Addition Solution Conc.:** 1000 (mmol L^{-1}). Concentration of catalyst solution being continuously added is 1000 mmol L^{-1} .
 - **Rate of Addition:** $50\text{E-}6$ (L min^{-1}). Rate of continuous addition of catalyst solution is $50 \times 10^6 \text{ L min}^{-1}$.
 - **After Time:** 5 (min). Continuous addition of catalyst solution commenced at 5 min.

CAKE Web App Guidance

Upload

System

Fitting Options

Data Manipulation

Output

Initial Reaction Volume

Continuous Volume Loss Rate

0.05

0.0

Est Rate Constant

Min Rate Constant

Max Rate Constant

0.003

Estimate * 1E-3

Estimate * 1E3

Reactant 1

Reactant 2

Catalyst

Product

☐ Use For Fitting

Species Name

Species Type

Stoichiometry

Catalyst

Catalyst

1

Initial Moles

Final Moles

Column

0

Specify Order Limits

Specify Poisoning Limits

Est Order

Min Order

Max Order

1

0

2

Continuous Addition

Discrete Addition

Addition Solution Conc.

Rate of Addition

After Time

1000

50E-6

5

+ Addition

- Addition

+ Species

- Species

Temperature Alteration

Discrete Subtraction

Fit

Download Fit

Output

Fit:

Upload

System

Fitting Options

Data Manipulation

Output

Rate Constants (k):

Fitted: 0.003

Error: 5.4e-10

Reaction Orders:

Fitted: 1, 1

Error: 3.9e-08, 1.8e-09

Quality of Fit Metrics:

RMSE: 5.174e-06 | MAE: 3.292e-06

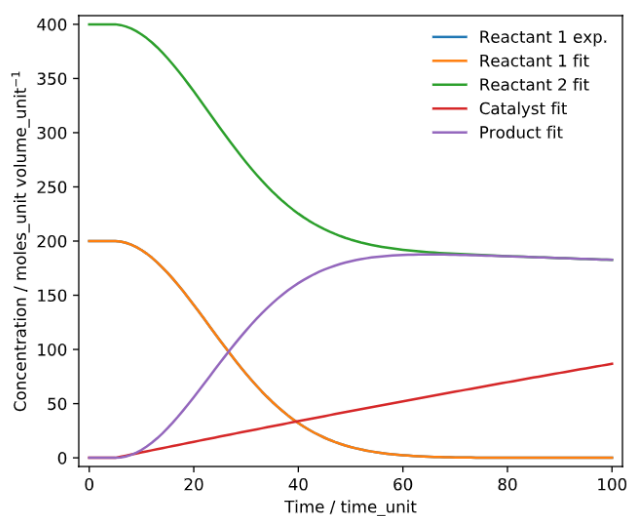
R²: 1 | Adjusted R²: 1

AIC: -2453 | BIC: -2445

Download file to see full results

Fit

Download Fit



Download Fit:

These results are available as “CAKE_example_fit.xlsx”.

- FitData:

CAKE Web App Guidance

	A	B	C	D	E	F
1	Time / min	Reactant 1 exp. conc	Reactant 1 fit conc. /	Reactant 2 fit conc. /	Catalyst fit conc. / m	Product fit conc. / mole
2	0	200	200	400	0	0
3	1	200	200	400	0	0
4	2	200	200	400	0	0
5	3	200	200	400	0	0
6	4	200	200	400	0	0
7	5	200	200	400	0	0
8	6	199.5009308	199.5009229	399.3011227	0.999000999	0.299276886
9	7	198.408371	198.4083625	398.0091609	1.996007984	1.192435897
10	8	196.7332611	196.7332561	396.1350507	2.991026919	2.668538502
11	9	194.4916382	194.491631	393.6948182	3.984063745	4.711556298
12	10	191.7043457	191.7043468	390.7093219	4.975124378	7.300628338
13	11	188.3967896	188.3967845	387.2039416	5.964214712	10.41037251
14	12	184.5984497	184.5984492	383.2081811	6.951340616	14.01128263
15	13	180.3425598	180.3425486	378.755247	7.936507937	18.07014984
16	14	175.6655121	175.6655113	373.8815668	8.919722498	22.55054416
17	15	170.6064758	170.6064676	368.6262696	9.900990099	27.41333437
18	16	165.2067261	165.2067139	363.0306506	10.88031652	32.61722281
19	17	159.5091705	159.5091577	357.1376162	11.85770751	38.11930077
20	18	153.557785	153.5577755	350.9911417	12.83316881	43.87559077
21	19	147.3970337	147.3970213	344.6356801	13.80670611	49.84163748
22	20	141.0714264	141.0714264	338.1157614	14.77832512	55.97290855
23	21	134.6247711	134.6247701	331.4751638	15.7480315	62.22562358
24	22	128.1000366	128.1000294	324.7568633	16.71583088	68.55680439
25	23	121.5385071	121.5385025	318.0022467	17.68172888	74.92506176

- OutputParams:

	A	B	C	D	E	F	G	H
1	Rate Constants	Rate Constant Errors	Reaction Orders	Reaction Order Errors	cies Poisons	Poisoning	RSS	RMSE
2	[0.00300001]	[5.41274183e-10]	[0.99999943 0.99999928]	[3.85253293e-08 1.79656238e-09]	N/A	N/A	2.7036781	5.1738854

- InputParams:

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	sim	t	xl	heet_name	spec_name	spec_type	vol0	stoich	mol0	mol_end	dd_sol_con	d_cont_ra	t_cont
2	False	(0.0, None <FileStorage	FitData		['Reactant ['r', 'r', 'c',	0.05		[1, 1, 1, 1]	[10.0, 20.0	[2.940561	[None, No	[None, No	[None, No

The reaction was determined to have a rate constant of $0.003 \text{ mmol L}^{-1} \text{ min}^{-1}$ and be first order in reactant and catalyst, as expected.